

# ELI — Capability Catalogue

ELI v2.1.73

August 2026

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## ELI Capability Catalogue — every action & module, what it actually does

**Purpose.** A systematic, ground-truth catalogue built by reading the real handlers and modules — not summarised from memory. It exists because conversational summaries of a 126k-LOC project keep undershooting; this is the persisted, exhaustive map. Built in committed batches.

**Method.** Action list comes from the live `capability_manifest.json` (**209** entries as of 2026-07-22; 190 routable + plugin-backed), verified against the `executor_enhanced.py` dispatch. **The always-current, auto-generated action list with activation phrases is `capabilities_and_actions.md`** — this catalogue is the deeper module-level read. Behaviour grounded in the handlers and the subsystem blueprints. Where a description is inferred from name+structure rather than a line-by-line handler read, it is marked (*inferred*).

**Status:** Batch 1 — action catalogue (this file will grow with the runtime/ + cognition/ module catalogues and the remaining unread files).

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### Headline finding: 208 (190 routable) is real but aliased

The manifest's 209 entries (190 routable) are honest (*measured* by `capability_sync`, not asserted) but inflated by **alias families** — multiple action names routing to one behaviour. Collapsed, there are roughly **~110 distinct capabilities**. Alias families are grouped below so the real surface is visible.

---

## 1. Conversation & reasoning

| Action(s)                    | What it does                                                                                       |
|------------------------------|----------------------------------------------------------------------------------------------------|
| CHAT                         | The default — full cognitive pipeline (router→agents→orchestrator→reasoning mode→governed output). |
| ANSWER, DIRECT_RESPONSE, SAY | Direct/short response surfaces (lighter than CHAT). ( <i>inferred for some</i> )                   |

| Action(s)               | What it does                                                                                                  |
|-------------------------|---------------------------------------------------------------------------------------------------------------|
| SET_AI_MODE             | Set the reasoning mode (Quick / Normal / Advanced / Research / Expert). Legacy names still accepted as input. |
| SEQUENCE, SEQUENCE_STEP | Multi-step action chaining (run a sequence of actions). ( <i>inferred</i> )                                   |
| TEMPLATE, NOOP          | Internal scaffolding / no-op.                                                                                 |

## 2. OS & application control

| Action(s)                                                                                                                                      | What it does                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OPEN_APP = OPEN_APPLICATION = LAUNCH_APP                                                                                                       | Launch an app, resolved via the <code>system_index</code> (your machine's executables — thousands, indexed live) + <code>portable_app_control</code> . On failure → <b>remediation offer</b> (real apt/snap/flatpak install on confirmation). |
| CLOSE_APP = CLOSE_APPLICATION = EXIT_APP = QUIT_APP                                                                                            | Close/quit an app.                                                                                                                                                                                                                            |
| FOCUS_APP, HIDE_APP                                                                                                                            | Focus / hide a window.                                                                                                                                                                                                                        |
| MINIMISE_APP = MINIMIZE_APP, MINIMISE_WINDOW = MINIMIZE_WINDOW, MAXIMISE_WINDOW                                                                | Window state control (UK/US spellings aliased).                                                                                                                                                                                               |
| MINIMISE_ALL, RESTORE_WINDOWS, TILE_WINDOWS, NEXT_WINDOW, PREVIOUS_WINDOW                                                                      | Desktop window management.                                                                                                                                                                                                                    |
| SWITCH_WORKSPACE                                                                                                                               | Switch virtual desktop/workspace.                                                                                                                                                                                                             |
| OPEN_SYSTEM_SETTINGS, OPEN_AUDIO_SETTINGS, OPEN_POWER_SETTINGS, OPEN_FILE_SYSTEM, OPEN_COMMUNICATION_HUB, OPEN_MEDIA_HUB, OPEN_NETWORK_BROWSER | Open specific OS setting panels / hub launchers.                                                                                                                                                                                              |
| OPEN_IDE, OPEN_IN_IDE                                                                                                                          | Open the IDE (optionally at a file).                                                                                                                                                                                                          |
| OPEN_BROWSER, OPEN_URL                                                                                                                         | Open browser / a URL (local hand-off; respects net toggle for fetches).                                                                                                                                                                       |

## 3. Input & screen control

| Action(s)                    | What it does                                                     |
|------------------------------|------------------------------------------------------------------|
| KEYBOARD                     | Send keypresses ( <code>os_controller.press_key</code> ).        |
| MOUSE_CONTROL                | Move/click the mouse ( <code>os_controller.mouse_click</code> ). |
| VOLUME                       | Get/set system volume.                                           |
| SET_CLIPBOARD, GET_CLIPBOARD | Read/write the clipboard.                                        |
| SCREENSHOT                   | Capture the screen.                                              |

## 4. Gaze (webcam eye-tracking)

| Action(s)                 | What it does                                                                       |
|---------------------------|------------------------------------------------------------------------------------|
| GAZE_ENABLE, GAZE_DISABLE | Start/stop the MediaPipe gaze engine (writes <code>latest_gaze.json</code> @10Hz). |
| GAZE_CALIBRATE            | Run gaze calibration.                                                              |
| GAZE_CLICK                | Move the cursor to where you're looking and click.                                 |

| Action(s)   | What it does              |
|-------------|---------------------------|
| GAZE_STATUS | Report gaze-engine state. |

## 5. Media

| Action(s)                                                                        | What it does                                                                                                                                                       |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PLAY_MEDIA                                                                       | Play a song/playlist on a named platform (Spotify playlist-tab / YouTube Mix-radio); honest about reachability; explicit platform never falls through to a second. |
| PAUSE_MEDIA, STOP_MEDIA, NEXT_MEDIA, PREVIOUS_MEDIA, REPEAT_MEDIA, SHUFFLE_MEDIA | MPRIS/playerctl transport control.                                                                                                                                 |
| MEDIA_CONTROL                                                                    | Generic media-control dispatch.                                                                                                                                    |
| SKIP_YOUTUBE_AD                                                                  | Skip a YouTube ad. ( <i>inferred</i> )                                                                                                                             |

## 6. Files & documents

| Action(s)                                                                        | What it does                                                                              |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| CREATE_FILE, CREATE_FOLDER, READ_FILE, LIST_DIR                                  | Filesystem ops (path-allowlist gated).                                                    |
| SUMMARIZE_FILE                                                                   | Summarise a file's contents (also the route for "read your persona.auto.txt / settings"). |
| CONVERT_DOCUMENT                                                                 | Convert document formats.                                                                 |
| CREATE_DOCUMENT = CREATE_DOC = DOC_GENERATE = GENERATE_DOCUMENT = WRITE_DOCUMENT | Generate a document (the Report-Builder family).                                          |
| ANALYZE_CSV, ANALYZE_PDF, ANALYZE_PDF_FOLDER, ANALYZE_IMAGE, OCR_IMAGE           | Local file-type analysers (CSV stats, PDF text/structure, image VL description, OCR).     |

## 7. Vision & screen understanding

| Action(s)           | What it does                                                        |
|---------------------|---------------------------------------------------------------------|
| ANALYZE_IMAGE       | Local GGUF vision-language description of an image.                 |
| OCR_IMAGE           | Tesseract OCR text extraction.                                      |
| AMBIENT_VISION      | Toggle periodic background screen glances.                          |
| SCREEN_LOCATE       | OCR-locate a named UI element on screen ("the button that says X"). |
| SCREEN_READ_ANALYZE | Screenshot → analyse what's on screen.                              |
| IMAGE_STATUS        | Report vision/image-engine state.                                   |

## 8. Coding & code-repair

| Action(s)                                                                   | What it does                                                                                                          |
|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| CODE_SOLVE                                                                  | The frontier coding agent: plan → DAG decompose → UCB tree search → verify (syntax/exec/tests) → repair → bug-memory. |
| GENERATE_SCRIPT = CREATE_SCRIPT = WRITE_SCRIPT = GENERATE_CODE = WRITE_CODE | Generate a script — routes through the coding agent (falls back to inline gen).                                       |

| Action(s)                                   | What it does                                                                                                    |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| GENERATE_PROJECT<br>EXAMINE_CODE            | Generate a multi-file project.<br>Tiered file scan (syntax/import → lint → gated LLM logic review) → offer fix. |
| FIX_FILE, CONFIRM_CODE_FIX, CANCEL_CODE_FIX | Apply/confirm/cancel a verified auto-reverting code fix.                                                        |
| FILE_AUDIT, SHOW_DIFF, CODE_CHANGES         | Audit a file / show a diff / report recent code changes.                                                        |

## 9. Self-management & self-improvement

| Action(s)                                           | What it does                                                                   |
|-----------------------------------------------------|--------------------------------------------------------------------------------|
| SELF_REPORT                                         | Grounded runtime self-report (deterministic, from live state).                 |
| SELF_ANALYZE, SELF_IMPROVE,<br>SELF_IMPROVEMENT_LOG | Analyse failures / surface improvement proposals / show the improvement log.   |
| SELF_PATCH                                          | Generate+apply a verified self-patch.                                          |
| SELF_TEST                                           | Run self-tests.                                                                |
| SELF_UPGRADE = SELF_UPDATE                          | Maintenance: git pull, pip, rebuild FAISS/KG, refresh manifest + system index. |

## 10. Introspection & audits (grounded, deterministic)

| Action(s)                                                                                                                                                 | What it does                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| RUNTIME_STATUS, RUNTIME_AUDIT                                                                                                                             | Live runtime report; RUNTIME_AUDIT also runs <b>health probes</b> (plugin mgr, memory, agent bus, habit integrity, recent failures).               |
| GUI_RUNTIME_AUDIT, IMPORT_AUDIT,<br>DIAGNOSE_WRAPPERS, RESOLVE_RUNTIME_PATHS<br>FRONTIER_STATUS, ELI_IDENTITY_AUDIT                                       | GUI/runtime audits, import smoke-audit, executor-wrapper diagnostic, path resolution.<br>Full cross-system matrix / identity-classification audit. |
| COGNITION_STATUS, EXPLAIN_COGNITION_RUNTIME,<br>EXPLAIN_MEMORY_RUNTIME<br>EXPLAIN_ALL_REASONING_MODES,<br>REASONING_MODE_STATUS                           | Cognition + memory runtime explainers (verbatim, deterministic).<br>Describe the 5 reasoning modes / current mode.                                 |
| EXPLAIN_LAST_RESPONSE, EXPLAIN_LAST_FAILURE,<br>NAME_SOURCE_AUDIT, ROUTING_FAULT_EXPLAIN<br>HARDWARE_PROFILE, GPU_STATUS, GET_STATUS,<br>AWARENESS_STATUS | Explain the last answer/failure, the source of your name, or a routing fault.<br>Hardware/GPU/general/awareness status.                            |
| HELP, LIST_CAPABILITIES                                                                                                                                   | Help / list the capability surface.                                                                                                                |

## 11. Memory & identity

| Action(s)                                                   | What it does                                                         |
|-------------------------------------------------------------|----------------------------------------------------------------------|
| MEMORY_RECALL, MEMORY_STORE, MEMORY_STATS,<br>MEMORY_STATUS | Recall/store memories; memory inventory/stats.                       |
| PERSONAL_MEMORY_SUMMARY,<br>PERSONAL_MEMORY_DEEP_EXPLAIN    | “What do you know about me” (clean) / deep memory-internals explain. |

| Action(s)                                                                                            | What it does                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| USER_IDENTITY_SUMMARY, USER_INFO_REPORT,<br>REFRESH_USER_INFO<br>SET_USER_NAME<br>MESSAGE_TIME_QUERY | Who-you-are summary / full profile report /<br>rebuild the living profile.<br>Set the user's name.<br>"What time did I first/last message (today)" —<br>from conversation_turns. |
| CLEAR_CHAT_HISTORY<br>PERSONA_LOCK_SET, PERSONA_LOCK_CLEAR,<br>PERSONA_LOCK_STATUS                   | Clear chat history.<br>Lock/unlock/inspect the persona.                                                                                                                          |

## 12. Habits, proactivity & goals

| Action(s)                                                                                          | What it does                                                                                                                |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| CONFIRM_HABIT, DECLINE_HABIT, HABIT_STATUS<br>MORNING_REPORT                                       | Approve/decline a detected habit; report habits.<br>The consolidated morning brief (news digest +<br>activity + attention). |
| PROACTIVE_START, PROACTIVE_STOP,<br>PROACTIVE_STATUS<br>EXECUTE_GOAL<br>BACKGROUND_JOBS, CHECK_JOB | Control the proactive daemon.<br><br>Execute a mission-layer goal (governed).<br>List background tasks / check a job id.    |

## 13. Self-healing remediation (Linux)

| Action(s)                                                                                                    | What it does                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PREPARE_REMEDIATION<br><br>CONFIRM_PENDING_REMEDIATION,<br>CANCEL_PENDING_REMEDIATION<br>CHECK_TARGET_STATUS | Diagnose a failure + build a repair plan<br>(e.g. install a missing app via apt/snap/flatpak).<br>Execute / cancel the pending repair (sudo<br>terminal, lock-handling, verification).<br>Check whether a target app/path is now present. |

## 14. Voice & transcription

| Action(s)                                                                                                                 | What it does                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DICTATE, TRANSCRIBE<br>LISTEN_FOR_COMMAND<br>STT_DIAGNOSTICS, VOICE_DIAGNOSTICS<br><br>SAY / SPEAK ( <i>plugin: tts</i> ) | Dictation / transcribe audio (faster-whisper).<br>One-shot listen.<br>Diagnose the STT/voice pipeline (mic, ducking,<br>wake gate).<br>Speak text (Piper TTS). |

## 15. Time, timers & utility

| Action(s)                                                                           | What it does                                                                                                                                       |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| TIME = GET_TIME, DATE = GET_DATE<br>SET_TIMER, SET_ALARM<br>CHECK_CHRONAL_ALIGNMENT | Current time / date.<br>Timers/alarms. ( <i>inferred</i> )<br>Easter-egg: "Chronal alignment nominal. Local<br>time: ..." (a playful time report). |

| Action(s)           | What it does                                                                                                   |
|---------------------|----------------------------------------------------------------------------------------------------------------|
| DATA_FABRICATOR     | Generate a document from a topic (via CREATE_DOCUMENT) <b>and open it in an editor</b> (code/gedit/kate/nano). |
| RUN_CMD, SHELL_EXEC | Run a shell command — <b>fail-closed</b> (blocked unless ELI_ALLOWED_CMDS is set or ELI Full Control is on).   |

## 16. Plugins (management)

| Action(s)                                                       | What it does                               |
|-----------------------------------------------------------------|--------------------------------------------|
| PLUGIN_LIST, PLUGIN_SEARCH, PLUGIN_STATUS                       | List installed / search registry / status. |
| PLUGIN_INSTALL, PLUGIN_UNINSTALL, PLUGIN_ENABLE, PLUGIN_DISABLE | Install/remove/enable/disable a plugin.    |

## 17. Plugin-backed capabilities (the 13)

| Action                                         | Plugin                 | What it does                                                                                                                                    |
|------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| WEB_SEARCH                                     | web                    | DuckDuckGo/SearXNG web search (toggle-gated).                                                                                                   |
| GET_WEATHER                                    | weather                | Local geocode + open-meteo (toggle-gated).                                                                                                      |
| NEW_NOTE/WRITE_NOTE, LIST_NOTES, SEARCH_NOTES  | notes                  | Markdown notes with FTS.                                                                                                                        |
| ADD_EVENT, LIST_EVENTS                         | calendar               | ICS calendar events.                                                                                                                            |
| POMODORO_START, POMODORO_STOP, POMODORO_STATUS | pomodoro               | Focus timers.                                                                                                                                   |
| CPU_USAGE, RAM_USAGE, SYSTEM_STATS             | system_stats           | CPU/RAM/disk/network.                                                                                                                           |
| SMART_HOME                                     | smart_home             | Smart-home control via ELI's own MQTT device server (ESPHome / Tasmota / Zigbee2MQTT); rooms, scenes, real automations. Home Assistant removed. |
| SPEAK                                          | tts                    | Text-to-speech.                                                                                                                                 |
| NEWS_FETCH                                     | (executor + news tool) | Fetch + synthesise news (rolling 3-hourly reflections → 24h digest).                                                                            |

(Also reachable: *web\_automation* plugin — Playwright browser navigate/search/screenshot; *document\_reader* plugin — PDF/docx read+index.)

## Reading note

Alias families that collapse the count: app-open (3), app-close (4), minimise (4 across window/app + UK/US), document-generate (5), script/code-generate (5), time/date (2 each), self-upgrade/update (2). Removing pure aliases yields **~110 genuinely distinct capabilities** — still an unusually broad surface

for a local single-user assistant, spanning OS control, media, files, vision, gaze, voice, coding, memory, introspection, autonomy, remediation, and plugins.

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## Part 2 — runtime/ module catalogue (80 files, ~22.6k LOC)

The largest package. It's the **grounding/governance + introspection + plumbing** layer that wraps the probabilistic model. Grouped by function:

### Grounding & anti-confabulation

| Module                                                        | LOC             | Role                                                                                                                                                                                                        |
|---------------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| deterministic_grounding_gate.py                               | 4301            | Renders control/status answers directly from live runtime (bypasses the model). 7 stacked render_action layers + an immutable policy engine (the active chain; one dead v10 fragment removed this session). |
| grounding_escalation.py                                       | 528             | When a <b>checkable factual</b> question is poorly grounded by the bus, escalates through agent tiers instead of letting the model confabulate (the "Eminem's real name" failure class).                    |
| diagnostic_patterns.py                                        | 112             | Regexes that catch vague/dynamic status confabulation ("currently processing updates...") and image-status fabrication.                                                                                     |
| control_contracts.py                                          | 1045            | Control-action evidence contract: build evidence → validate the model's output doesn't violate it → finalise.                                                                                               |
| contracts/grounded_control.py,<br>contracts/runtime_status.py | (in contracts/) | Runtime-status question detection, live-evidence build, repair/validate.                                                                                                                                    |
| memory_evidence.py                                            | 221             | Collects memory evidence for grounding a turn.                                                                                                                                                              |
| persistence_gate.py                                           | 188             | Gates what gets stored — refuses to persist internal dumps / error-pattern noise as memory.                                                                                                                 |

### Self-honesty / introspection / self-reporting

| Module                | LOC | Role                                                                                                              |
|-----------------------|-----|-------------------------------------------------------------------------------------------------------------------|
| live_introspection.py | 730 | Live runtime snapshot, last trace, stored user name, mines user-fact candidates, agents-for-action, build_report. |



| Module                         | LOC | Role                                                                                                                                      |
|--------------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------|
| deterministic_introspection.py | 573 | The engine's live diagnostic dispatcher (handle_diagnostic_action) — deterministic answers for RUNTIME_STATUS / EXPLAIN_* / IMPORT_AUDIT. |
| truth_report.py                | 301 | Runtime truth report (git, nvidia, GGUF runtime, import health).                                                                          |
| frontier_status.py             | 423 | Full cross-system status matrix (run-time/memory/awareness/proactive/image/world)                                                         |
| eli_identity_audit.py          | 416 | Identity-classification audit (source inventory, contract counts, capability matrix).                                                     |
| reasoning_status.py            | 201 | Current reasoning-mode reporting.                                                                                                         |
| experimental_inventory.py      | 146 | Inventory of experimental projects.                                                                                                       |

## Self-improvement & code-awareness

| Module                    | LOC  | Role                                                                                                                                                |
|---------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| self_improvement.py       | 1296 | Failure logging/clustering, patch generate→verify→apply→ <b>auto-revert</b> , full patch cycle, plugin-stub gen.                                    |
| code_examiner.py          | 802  | Tiered file error scan (syntax/import → lint → gated LLM) → offer → verified fix.                                                                   |
| code_monitor.py           | 262  | Detects source changes via git diff, classifies by subsystem, summarises for memory/context (ELI is aware of its own code changes).                 |
| capability_sync.py        | 392  | <b>AST-discovers</b> the live capability surface, diffs, writes capability_manifest.json — this is why the count is <i>measured</i> , not asserted. |
| generated_script_guard.py | 1082 | Validates LLM-generated scripts, <b>quarantines invalid ones</b> , and ships vetted canned scripts for known patterns (GPU-watch, redshift, etc.).  |
| repair_policy.py          | 23   | Policy line: proposal layers vs source-mutation layers.                                                                                             |

## Self-healing remediation

| Module                  | LOC  | Role                                                                                                                                                                               |
|-------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| grounded_remediation.py | 1605 | Diagnoses failures (missing app/path/browser) → builds an apt/snap/flatpak repair plan → offers → <b>executes</b> (sudo terminal, lock-handling, verify). Writes incident records. |
| incident_log.py         | 21   |                                                                                                                                                                                    |

## Awareness & boot

| Module               | LOC | Role                                                                                                            |
|----------------------|-----|-----------------------------------------------------------------------------------------------------------------|
| awareness_boot.py    | 297 | Boots all awareness subsystems at startup, returns an AwarenessState the engine queries.                        |
| action_commitment.py | 168 | Detects when ELI's reply COMMITS to an action (so the pipeline re-runs and actually does it — no fake actions). |

## Autonomy / operator (governed)

| Module                                                                                         | LOC         | Role                                                                                                                    |
|------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------|
| operator_state.py,<br>operator_feed.py,<br>operator_feed_normalized.py,<br>pending_proposal.py | ~220<br>142 | Operator console state: proposals, goals, self-model status, event feed.<br>Pending-proposal state (extract/set/clear). |
| approval_engine.py                                                                             | 103         | Who may propose / evaluate a proposal record (governance).                                                              |

## Response surfaces & governance

| Module                                                                              | LOC  | Role                                                                                                                |
|-------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------|
| user_visible_response_surface.py                                                    | 336  | Installs the engine's user-visible response surface (runtime/identity/name-source formatting + streaming coercion). |
| visible_output.py,<br>visible_text.py,<br>output_sanitizer.py                       | ~200 | Central visible-output contract; stringify/sanitise streamed output.                                                |
| response_policy.py,<br>response_contracts.py,<br>response_packets.py                | ~205 | Classify response mode; per-action contracts; final-answer request packets.                                         |
| final_response_assembly.py,<br>final_response_provider.py,<br>fastpath_responder.py | ~175 | Assemble the final prompt; per-action generation decoration; fastpath context.                                      |

## Personal-memory surfaces

| Module                                       | LOC  | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| personal_memory_surface.py                   | 431  | "Is this a personal-memory query" + surface builder.<br>Clean "what do you know about me" report (reset-aware, poison-filtered, dynamic-fact aging).<br>Deep memory-internals explain (schema/tables/functions) + routing-fault explain.<br>Extracts user facts from turns (role/interests/field/"remember that I..."), writes user_patterns + LLM session summaries; recency refresh.<br>Validate identity candidates; persona/identity lock state. |
| personal_memory_clean_response.py            | 314  |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| personal_memory_deep_response.py             | 453  |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| profile_extractor.py                         | 740  |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| identity_validation.py,<br>identity_guard.py | ~240 |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## Typed pipeline plumbing (evidence/packets)

| Module                                                                                                                                                                                                                    | LOC  | Role                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| evidence_ledger.py                                                                                                                                                                                                        | 595  | Records artifacts/events with signatures; recent generated artifacts; status evidence.<br>Scores competing evidence (stage packets + tool results + goals), dedup-by-fingerprint, keep-max.<br>The typed packet substrate: route/plan/evidence/generation/output packets flow between stages; this is the plumbing behind "no fake actions". |
| evidence_arbitration.py                                                                                                                                                                                                   | 195  |                                                                                                                                                                                                                                                                                                                                              |
| evidence_store.py,<br>stage_packet_store.py,<br>stage_packets.py,<br>pipeline_models.py,<br>retrieval_packets.py,<br>typed_stage_bridge.py,<br>packet_native_downstream.py,<br>single_pass_authority.py,<br>tool_result_* | ~700 |                                                                                                                                                                                                                                                                                                                                              |
| background_tasks.py                                                                                                                                                                                                       | 253  | In-process multi-threaded task manager (heavy work → job id → CHECK_JOB).<br>Per-turn budgets/timeouts/context size from runtime snapshot.                                                                                                                                                                                                   |
| runtime_policy.py                                                                                                                                                                                                         | 76   |                                                                                                                                                                                                                                                                                                                                              |

## Security

| Module      | LOC | Role                                                              |
|-------------|-----|-------------------------------------------------------------------|
| security.py | 210 | SecurityManager — fail-closed command/path/app allowlist sandbox. |

## Part 3 — cognition/ module catalogue (27 files, ~14.0k LOC)

The thinking layer: agents, orchestration, inference, persona, reasoning, governance.

### Orchestration & agents

| Module                 | LOC  | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| agent_bus.py           | 3276 | 15 specialist agents on a dependency DAG (topological layers) + calibrated weight-free confidence aggregation + per-action agent selection.<br>The 12-stage retrieval pipeline: HyDE → FAISS + FTS5 + KG-BFS + RAG → hybrid merge → heuristic rerank (lexical × recency × importance; cross-encoder is the designed upgrade) → context assembly.<br>Hypothetical-document-embedding query expansion.<br>Candidate reranking (token overlap + source priority).<br>Wraps introspection for the bus (pipeline/memory/runtime/audit).<br>LLM intent parsing fallback (GGUF, cached). |
| orchestrator.py        | 928  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| hyde.py                | 69   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| reranker.py            | 79   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| introspection_agent.py | 173  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| llm_intent.py          | 251  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

### Inference

| Module              | LOC  | Role                                                                                                                                                                                                                                                                                          |
|---------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gguf_inference.py   | 2803 | Model-agnostic GGUF inference: model resolution (no baked model), family-aware chat templating, graceful GPU-layer fallback, streaming, output cleaning, token budgeting.<br>Thin GGUF broker (infer) used by agents/coding/patching.<br>Chat response + streaming helpers, turn persistence. |
| inference_broker.py | 215  |                                                                                                                                                                                                                                                                                               |
| chat_model.py       | 292  |                                                                                                                                                                                                                                                                                               |

## Reasoning & engagement

| Module                | LOC | Role                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reasoning_modes.py    | 631 | The 5 modes — canonicalisation, per-mode private system instruction, execution contract (samples/branches/stages + dynamic token budget), reasoning-leak stripping. Session depth tracking → <b>auto-escalates reasoning mode</b> (Quick→Normal→Advanced→Research) as a conversation deepens; session narrative. Turn-scoped pinned facts (pin/absorb/evict/persist/restore). |
| engagement_tracker.py | 249 |                                                                                                                                                                                                                                                                                                                                                                               |
| working_memory.py     | 325 |                                                                                                                                                                                                                                                                                                                                                                               |

## Context & grounding

| Module                 | LOC | Role                                                                                                                                                                                                                                                             |
|------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| context_synthesiser.py | 567 | Builds the precise prompt context: persona handoff, turns block, vector block, live-runtime brief, budgeting. Lighter persona+memory context builder + fallback guard. Identity + memory-inventory rendered directly from profile/DBs (direct grounded answers). |
| context_builder.py     | 118 |                                                                                                                                                                                                                                                                  |
| grounded_status.py     | 654 |                                                                                                                                                                                                                                                                  |

## Persona (the living voice)

| Module                                  | LOC  | Role                                                                                                                                                                                                                                                                          |
|-----------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| persona.py                              | 375  | Canonical persona authority — base + auto sections, preferences, compose/refresh. Re-derives the persona overlay from memory/reflection/habits/runtime patterns; KG population; stale-fact aging. Cleans/dedups/prunes the auto-persona. Persona status report; values store. |
| persona_updater.py                      | 746  |                                                                                                                                                                                                                                                                               |
| persona_hygiene.py                      | 131  |                                                                                                                                                                                                                                                                               |
| persona_status.py,<br>persona_values.py | ~108 |                                                                                                                                                                                                                                                                               |

## Output governance (consolidated this session)

| Module                                           | LOC   | Role                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| output_governor.py                               | 1311  | <b>Canonical</b> governance: sanitize, role-prefix/identity-drift repair, self→user confusion repair, style cleanup, confabulation detection, quality scoring, memory-worthiness, GGUF-artifact cleaning (clean_gguf_artifacts), evidence validation.<br><b>Re-export shims</b> → output_governor (kept for back-compat).<br>Analyses recent user turns → tone preferences ELI adapts to over time. |
| response_governance.py,<br>response_sanitizer.py | 28+14 |                                                                                                                                                                                                                                                                                                                                                                                                     |
| tone_analyzer.py                                 | 373   |                                                                                                                                                                                                                                                                                                                                                                                                     |

## Profile

| Module               | LOC | Role                                                                                                     |
|----------------------|-----|----------------------------------------------------------------------------------------------------------|
| user_info_builder.py | 755 | The living, versioned user profile: multi-source gather, noise filter, categorise, hash, diff-on-change. |

## Part 4 — remaining packages (module catalogue)

### kernel/ (the engine + boot) — 14.5k LOC

| Module         | LOC   | Role                                                                                                                                                                                                                                                                                                     |
|----------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| engine.py      | 13841 | CognitiveEngine — the conductor: persona, generation settings, the 5 _run_* reasoning passes, grounding overrides, synthesis prompt build + the context-bloat cap, fragment/placeholder guards, dispatch gate to bus vs orchestrator, startup loops (reflection/habit/scheduler/self-improve/proactive). |
| world_model.py | 252   | Symbolic self-model: Identity/Runtime/Memory/Goal/Capability states + snapshot/merge.                                                                                                                                                                                                                    |

| Module                   | LOC  | Role                                                                                 |
|--------------------------|------|--------------------------------------------------------------------------------------|
| state.py                 | 367  | User/runtime state + profile (active user id, name, profile text).                   |
| self_upgrade.py          | 535  | Self-upgrade orchestrator (git pull, pip, rebuild FAISS/KG, manifest, system index). |
| scheduler.py             | 74   | Kernel thread-pool/timer scheduler (generic).                                        |
| pipeline.py, task_bus.py | ~180 | Pipeline step description; task bus.                                                 |

### core/ (paths, settings, hardware, safety) — 6.6k LOC

| Module                        | LOC  | Role                                                                                                                                      |
|-------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------|
| runtime_settings.py           | 1135 | Canonical settings.json load/save, legacy-key migration, portable-path healing/sanitising.                                                |
| hardware_profile.py           | 1302 | Auto-detect GPU/VRAM/RAM → fit ctx/gpu_layers/batch (KV-cache + compute-buffer reserve); model discovery; hardware authority enforcement. |
| paths.py                      | 601  | Single-source-of-truth path resolution (data/config/cache/db/models/voices...).                                                           |
| startup_hardware_optimizer.py | 655  | Boot-time hardware optimiser (GPU select, layer/ctx allocation, mode presets).                                                            |
| config.py                     | 350  | Thin config shim over runtime_settings (canonical key mapping).                                                                           |
| model_download.py             | 674  | Curated GGUF catalogue + VRAM-based recommend + download.                                                                                 |
| dag.py                        | 474  | <b>Generic DAG engine</b> (Kahn topo-order, parallel layers, critical path) — shared by the agent bus + coding engine.                    |
| dynamic_runtime_budget.py     | 215  | Per-boot runtime budget derivation.                                                                                                       |
| netguard.py                   | 451  | <b>Offline-by-default</b> socket-level network gate (fail-closed) + guarded_urlopen/http_get_json.                                        |
| memory_reset.py               | 329  | Factory-reset of memory/identity (scrub names, clear DBs, backup).                                                                        |
| cognition_tunables.py         | 269  | User-tunable knowledge-gathering limits + synthesis cap registry (GUI-surfaced).                                                          |

| Module                                                                                                               | LOC   | Role                                                                                            |
|----------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------|
| grounding.py                                                                                                         | 147   | is_grounded_query classifier.<br>STT-robust self-harm detector +<br>persona steering directive. |
| crisis_guard.py                                                                                                      | 111   |                                                                                                 |
| portable_paths.py, db_paths.py,<br>legacy_paths.py, first_run*.py,<br>compatibility.py,<br>architecture_contracts.py | small | path helpers, first-run, compat,<br>ownership map.                                              |

## memory/ — 7.0k LOC

| Module                                                                                                               | LOC   | Role                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| memory.py                                                                                                            | 4734  | The Memory god-class (~50 methods): semantic store/recall, conversations, habits, failures/improvements, weight decay, KG bridge, mark_failure_resolved, disable_invalid_habit_rules. |
| knowledge_graph.py                                                                                                   | 643   | Entity/relation graph + multi-hop BFS (related) + context_for_prompt + extract-from-memory.                                                                                           |
| habits_memory_db.py                                                                                                  | 470   | Habit rules/events store + cheap embed/recall.                                                                                                                                        |
| vector_store.py                                                                                                      | 430   | FAISS index (L2, 1/(1+dist) sim) + nomic embedder + keyword fallback + auto-rebuild.                                                                                                  |
| system_index.py                                                                                                      | 278   | OS app/exe/dir index (the launcher backing — your machine's executables, thousands, indexed live per machine).                                                                        |
| memory_truth.py,<br>memory_adapter.py,<br>memory_service.py, stores.py,<br>sqlite_memory.py,<br>populate_memories.py | small | inspection/compat/session helpers.                                                                                                                                                    |

## perception/ — 6.8k LOC

| Module       | LOC  | Role                                                                                                                                               |
|--------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| audio_stt.py | 1978 | faster-whisper STT + VoiceGate (wake-word, debounce, incomplete-command wait), self-echo suppression, output ducking, per-user voice profile bias. |



| Module                                                                                                                | LOC   | Role                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------|
| vision.py                                                                                                             | 692   | Local GGUF VL (Moondream fast / primary), hot-swap with text model, CPU-pinned CLIP, OCR. |
| tts_router.py                                                                                                         | 1128  | Piper/espeak TTS, voice selection, unspeakable-fragment guard.                            |
| os_controller.py                                                                                                      | 573   | Screenshot/volume/keys/mouse/clipboard + gaze_click.                                      |
| screen_locator.py                                                                                                     | 410   | OCR (tesseract) → locate a named UI element on screen.                                    |
| gaze_engine.py                                                                                                        | 358   | MediaPipe face-gaze + calibration mapper + One-Euro smoothing → latest_gaze.json @10Hz.   |
| local_whisper_stt.py,<br>ambient_vision.py, analyze_{pdfs,image,mesh,csv}.py,<br>log_rotation.py,<br>voice_worker*.py | ~1.2k | Whisper backend; ambient glances; file analysers; log rotation; voice workers.            |

### planning/ (autonomy, habits, proactive) — 3.9k LOC

| Module                                                                                                      | LOC  | Role                                                                                               |
|-------------------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------|
| proactive_daemon.py                                                                                         | 1459 | Background 10-min loop: pattern/code analysis, habit detect+offer, morning report, error tracking. |
| autonomy_scheduler.py                                                                                       | 212  | Policy-gated (observe/proposal/goal-driven) goal scheduler w/ cooldown + attention queue.          |
| habits.py                                                                                                   | 344  | Habit detection (timestamp→HH:MM clustering), offer/pending state, disabled-by-default.            |
| habits_scheduler.py                                                                                         | 152  | Fires active habits at their time (self-heals legacy rows, once-per-minute dedupe).                |
| goal_store.py, goal_models.py, goal_tick.py, operator_goal_actions.py                                       | ~400 | Mission goals (priority/cadence/risk/constraints/success-criteria) → governed proposals.           |
| attention_queue.py, proposal_queue.py, proposal_*.py, jobqueue*.py, autonomy_controller.py, task_planner.py | ~900 | Attention ranking; proposal queue/archive; job queue; safe autonomy ticks.                         |

### coding/ (the frontier coder) — 2.0k LOC

| Module              | LOC  | Role                                                                           |
|---------------------|------|--------------------------------------------------------------------------------|
| agent.py            | 215  | CodeAgent.solve — composes the loop; broker-backed generator; full provenance. |
| bug_memory.py       | 340  | Bug classification + (signature→fix) long-term SQLite memory w/ fuzzy recall.  |
| verification.py     | 310  | Gate ladder (syntax→exec→tests) + weighted scoring + test synthesis.           |
| sandbox.py          | 227  | Bounded isolated execution (scrubbed env, CPU limit, timeout).                 |
| search.py           | 176  | UCB1 tree search over a temperature-ladder beam.                               |
| plan_graph.py       | 164  | Subtask DAG decompose + topological solve + compose.                           |
| planner.py, cost.py | ~210 | Planner/implementer split; cost → foreground/background decision.              |

### learning/ (real LoRA self-training) — 3.3k LOC

| Module                                                                                                                                                            | LOC   | Role                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------|
| lora_trainer.py                                                                                                                                                   | 656   | Real torch/PEFT/transformers Trainer LoRA run.                                  |
| lora_trainer_guard.py                                                                                                                                             | 571   | TrainerTarget plans + safety validation.                                        |
| dataset_builder.py                                                                                                                                                | 558   | Build supervised dataset from logged turns/corrections (PII-redacted, deduped). |
| lora_eval.py                                                                                                                                                      | 491   | Adapter eval suite.                                                             |
| bootstrap_phi3_base.py,<br>base_model_resolver.py,<br>dataset_filters.py,<br>export_trainable_dataset.py,<br>merge_reviewed_datasets.py,<br>training_preflight.py | ~1.4k | Trainable base download/resolve, quality gates, export/merge, preflight.        |

### contracts/, integrations/, system/, utils/, cli/

| Module                        | LOC | Role                                                                                                |
|-------------------------------|-----|-----------------------------------------------------------------------------------------------------|
| contracts/runtime_status.py   | 471 | Canonical runtime-status evidence contract (detect question, build live evidence, validate/repair). |
| contracts/grounded_control.py | 210 | Grounded-control synthesis guard (evidence-complete? suppress bad clarifications).                  |

| Module                                   | LOC  | Role                                                                                                                                                                                                                                                                                                    |
|------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| integrations/mpiris/playerctl_backend.py | 662  | MPRIS2/playerctl media backend (player resolve, transport, status).<br>Optional Ollama backend client.<br>Cross-platform installed-app discovery + open/close/minimise.<br>Cross-platform layer (open url/file/app, volume, clipboard, notify, executables).<br>Central logger; headless terminal REPL. |
| integrations/ollama/client.py            | 517  |                                                                                                                                                                                                                                                                                                         |
| system/portable_app_control.py           | 338  |                                                                                                                                                                                                                                                                                                         |
| utils/platform_compat.py                 | 1053 |                                                                                                                                                                                                                                                                                                         |
| utils/log.py, cli/headless.py            | ~200 |                                                                                                                                                                                                                                                                                                         |

### world/ (embodied self-model) — ~1.2k LOC

| Module                                                                                    | LOC   | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| agency/autonomy_engine.py                                                                 | 267   | Awareness vector (easing/decay) → avatar-room routing + symbolic-object creation during reasoning;<br>provenance/snapshots/journal.<br>Consistency classes; goal decay; default habits; reflection→world; world identity.<br>World state schemas (rooms/objects/events/actions/awareness); object templates.<br>The PySide6 “Eli’s World” tab (house/rooms/avatar/objects, live state).<br>Avatar behaviour/locomotion/persona-map; journal/provenance/snapshots/storage; event bus + bridge. |
| agency/{policy,goal_ecology,habits_engine,reflection_bridge,world_consistency_classes.py} | small |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| core/schemas.py,<br>core/ontology.py                                                      | ~160  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| renderers/pyside6/{world_scene,world_panel}.py                                            | ~195  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| avatar/*, persistence/*,<br>world_event_bus.py,<br>local_world_bridge.py                  | ~700  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

### gui/ (PySide6 desktop) — ~20.3k LOC

| Module                    | LOC   | Role                                                                                                                                                                 |
|---------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eli_pro_audio_gui_v2_0.py | 12100 | Main window: 12 tabs + adapters (CentralMemory/LocalModel/Ollama/Executor bridges, the _GUIEngineAdapter), chat, drag-drop, reasoning-mode auto-select, all toggles. |

| Module                                                                                  | LOC  | Role                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| labs_tab.py                                                                             | 5694 | Labs workspace: Notebook, Memory browser, Jupyter launcher, Calculator(+constants), Physics tables, <b>Report Builder</b> (evidence-grounded docs), File-Chat, Workspaces, Sim-IDE. Launcher / first-boot auto-tune / main(). |
| app.py                                                                                  | 817  |                                                                                                                                                                                                                               |
| panels/startup.py                                                                       | 1305 |                                                                                                                                                                                                                               |
| panels/settings.py                                                                      | 723  | Advanced Settings (Agents/Models/ <b>Cognition</b> /Plugins/Self-Upgrade).                                                                                                                                                    |
| docks/operator_console_dock.py                                                          | 303  | Governed operator console (proposals/goals/policy/attention).                                                                                                                                                                 |
| widgets/, tabs/,<br>panels/agent_wizard.py,<br>docks/proactive_dock.py,<br>qt_compat.py | ~1k  | Ollama selector, experimental/world tabs, agent wizard, proactive dock, Qt shim.                                                                                                                                              |

### plugins/ — 10 built-ins + manager

| Module                                                                                             | Role                                                                            |
|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| manager.py (553L)                                                                                  | Discover/install/enable/disable/execute; auto-load; builtin-stub gen; registry. |
| web, web_automation, weather, calendar, notes, pomodoro, system_stats, media, tts, document_reader | The 10 built-in plugins (see Part 1 §17).                                       |
| base/base.py                                                                                       | Plugin base class + action validation + loader.                                 |

## Part 5 — verified engine request lifecycle (engine.process, line 8219)

Read end-to-end this session. The real entry flow, in order:

1. **Pipeline trace id** assigned (ELI\_PIPELINE\_TRACE for observability).
2. **Prompt-injection guard FIRST** — `_eli_sanitize_user_input` runs before the router or LLM ever sees the raw text, so injected instructions can't reach them unsanitised. (A security control that sits at the very top of the pipe.)
3. **Multi-question splitter** (kernel/engine.py, always on) — if a message has >1 ? **and** >25 words, it splits into standalone sub-questions (cap 4) and answers each; a second pass splits single-? compounds like "who are you, and who am I". So genuinely multi-part asks get multiple grounded answers.
4. Router → agent bus / orchestrator gate → reasoning mode → executor → governed output (Parts 1-4).

Note: engine.py carries **51 PHASE... markers** — the same "added beside, not folded in" accretion as the grounding gate. It works and is well-guarded, but the interior is layered patches; that residue (engine 13.4k / executor 14.3k / GUI 11.0k) is documented here at behavioural level, not line-by-line.

---

## Part 6 — reasoning modes & autonomous deepening

The 5 reasoning modes were renamed (public layer; internal keys unchanged) and made adaptive + self-deepening.

| Public mode     | Internal key      | Multi-pass algorithm      | Agent time budget | Deepen iters | Confidence target |
|-----------------|-------------------|---------------------------|-------------------|--------------|-------------------|
| <b>Quick</b>    | quick             | single pass               | 1.0×              | 0 (never)    | 0.45              |
| <b>Normal</b>   | chain_of_thought  | one reasoned pass         | 1.0×              | 1            | 0.55              |
| <b>Advanced</b> | self_consistency  | N samples + consensus     | 1.5×              | 2            | 0.65              |
| <b>Research</b> | tree_of_thoughts  | branch + prune            | 2.0×              | 3            | 0.75              |
| <b>Expert</b>   | constitutional_ai | draft → critique → revise | 2.5×              | 4            | 0.80              |

**Auto-selection** (cognition/engagement\_tracker.py): the mode is escalated automatically as a conversation deepens (Quick→Normal→Advanced→Research) without the user asking. (Internal strategy keys are stable; users only ever see the public names.)

**Per-mode agent time budgets** (reasoning\_modes.mode\_budget\_multiplier, Stage 1b): each mode scales every agent’s timeout — quick/normal unchanged, deeper modes get proportionally more time. Threaded via intent["\_mode\_budget\_mult"] → agent\_bus.\_collect\_layer. Tunable in the Cognition tab (cog.mode\_budget\_\*).

**Confidence-driven iterative deepening** (runtime/grounding\_escalation.escalate, Stage 2): on a poorly-grounded *checkable-factual* turn, ELI re-dispatches the agent bus broadly, **escalating the reasoning mode one tier per iteration** and (Stage 3a) **raising the gather budget** each pass (1.5×→2.0×→2.5×), until grounding crosses the per-mode target or the per-mode iteration budget is spent (early no-improvement break). Banter/opinion/command/meta never deepen.

**Background deepening** (runtime/background\_deepening.py, Stage 3b): a Quick answer returns instantly; if it was a poorly-grounded factual turn, a background thread keeps gathering (broad re-dispatch at Advanced + 2× gather) and surfaces a better answer in the Proactive panel **only if** it crosses grounding 0.55 and isn’t degenerate. Tightly gated (quick-only, grounding<0.40, factual-only, deduped, 1 in-flight, 10-min cooldown). Toggle: cog.background\_deepen / ELI\_BACKGROUND\_DEEPEN.